

Analysis of Emotional Tendency and Syntactic Properties of VR Game Reviews

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ABSTRACT

The studies of player reviews can help game developers design and optimize VR games. To this end, we investigated 288,685 reviews from 506 VR games on the Steam platform to analyze their sentiment tendencies using the machine learning-based model SKEP, which finds that although some of the reviews are “recommend”, they actually have opposite emotional tendencies. We also study the syntactic properties based on the natural language processing (NLP) kits Stanza and NLTK library, and we find that cybersickness is a significant concern for players.

Index Terms: Applied computing—Computers in other domains—Personal computers and PC applications—Computer games; Human-centered computing—Human-computer interaction (HCI)—HCI design and evaluation methods—User studies;

1 INTRODUCTION

With people’s increasing demand for diversified forms of entertainment, VR games are becoming more and more popular. However, creating an enjoyable VR game is not an easy task. In-depth and comprehensive analysis of players’ reviews has essential value for designing and developing a charming VR game. In this paper, we have chosen 506 VR games on Steam and analyzed 288,685 player reviews, using NLP-based models to study the sentiment tendencies and representative words in the reviews to understand the emotion and concern of players. In this way, we can help developers get valuable feedback from players’ experiences, thus bettering them to design and develop high-quality VR games.

The salient contributions of this paper can be summarized as follows.

- We study sentiment tendencies in game reviews and analyze the connections and differences between sentiment tendencies and recommendation attitudes.
- We analyze high-frequency words in reviews to understand how players describe specific characteristics of VR games.

2 BACKGROUND

Popular game distribution platforms allow players to post personal reviews after experiencing games, then their reviews will be displayed to other consumers on the main page of the game [1, 5, 11]. Steam [10] is the most famous video game distribution platform in the world. The Steam platform suggests players post their overall attitude about the game, either “recommend” or “not recommend”. The “recommend” review rate and the number of reviews of the game will be shown on the game’s main page to provide a reference for consumers.

While it’s a nascent industry, there’s already plenty of researches on VR game reviews [2–4]. It’s not hard to see from previous

research that analyzing VR games’ reviews can significantly help developers extract more concrete flaws and expectations.

3 EXPERIMENT

We use a Python web crawler to extract information of VR-exclusive games on Steam. Each review includes a player’s recommendation attitude, the number of reviews the player ever wrote, the total playtime, the detailed review text, et al. We select 506 VR games with no less than 50 reviews for each game, and the total number of reviews for these VR games is 327,530. Among all the reviews, 14,853 (5%) are written in Chinese, and 286,222 reviews (87%) are written in English.

Assisted by the Sentiment Knowledge Enhanced Pre-training (SKEP) model [8], we analyze the sentiment tendency of both English and Chinese players’ reviews after eliminating invalid reviews.

For the English reviews, our review tagging depends on the algorithm proposed by Toutanova et al. [9]. The Chinese part of speech tagging uses the Lexical Analysis of Chinese tool. Different forms of words are converted to their original format, providing a unified format for the following word frequency statistics. At last, we take NLTK [6] to give the interface of lemmatization and part of speech tagging. After speech tagging, we use word frequency statistics to identify the words with higher frequency in different parts of speech. We use the augmented Bi-LSTM-based deep biaffine neural dependency parser of Stanza [7] for dependency parsing, tackle dependency analysis to study the adjectives semantic of specific keywords.

4 EXPERIMENTS RESULTS

4.1 Sentiment Analysis of VR Game Reviews

The sentiment analysis is conducted on 14,123 Chinese reviews and 274,562 English reviews.

4.1.1 Recommendation Attitude of Reviews

Recommendation attitude is the most important attribute of player reviews with only two choices that can be selected, “recommend” or “not recommend”. Among the 288,685 comments, there were 257,977 “recommend” reviews (89%) and 30,708 “not recommend” reviews (11%). Most players tend to give favorable and supportive reviews for VR games.

4.1.2 Relationships between Sentiment Tendency and Recommendation Attitude

Among the “recommend” reviews of English reviewers, the number of reviews with a positive sentiment tendency is 212,350 (87%), and the number of reviews with a negative tendency of sentiment is 33,140 (13%). Among the “not recommend” reviews, there are 660 reviews (2%) with a positive sentiment tendency and 28,412 reviews (98%) with a negative sentiment tendency. Among the “recommend” reviews in Chinese, the number of reviews with a positive sentiment tendency is 9,335 (75%), and the number of reviews with a negative tendency of sentiment is 3,152 (25%). Among the reviews with the “not recommend” attitudes, 159 reviews (10%) possessed a positive sentiment tendency, and 1,477 reviews (90%) possess a negative tendency of sentiment.

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In general, the emotional tendency of players is consistent with their recommendation attitude. However, there are also differences in the proportion of sentiment tendency between Chinese and English reviews. The proportion of English reviews with the same emotional tendency and recommendation choice is 88%, while this proportion in Chinese reviews is only 77%. It may be interpreted that Chinese speaking players tend to express their opinions more implicitly. Game developers targeting at Chinese market may attach more attention to this difference.

Looking at all the reviews as a whole, we can spot that even some players would like to show their support or encouragement to the game, they still have strict requirements and high expectations. They should be the very customers that the developers of VR games need to pay special attention to. It is also important to find negative information or weakness about VR games in these “recommend” reviews. Studying these reviews will also help VR game developers find out existing problems and improve their games.

4.2 Word Analysis of All VR Game Reviews

Since word analysis has no significant differences between English and Chinese reviews, we concentrate fully on English. Word frequency statistics are carried out for three categories: nouns, verbs and adjectives. Then we rank each category of words according to the order of frequency from high to low.

Verbs analysis. Verbs in the reviews are mainly related to actions and movement in VR games, such as “run”, “move”, “shoot”, etc. They reflect the importance that players attached to game control experiences. Verbs that relate to the expectations of the game indicate the directions that players wish the game would continue to be updated and improved. It can be a driving force for developers to make and improve their games.

Nouns analysis. Nouns that relate to VR equipment and configuration have high frequencies, which shows that players attach importance to the hardware equipment and space required by VR games. In addition, “immersion” also has a high rate, indicating that players focus on the realistic experience of immersive atmospheres in VR games.

After combining several nouns for vertigo, we find that the symptom of cybersickness accounted for a relatively high proportion, indicating that it is also a vital flaw that plagues VR game players.

Adjectives analysis. For adjectives, the word “real” is at the top of the frequency ranking, indicating that the sense of reality in VR games is of great concern to players. Also, the adjective “smooth” is prominent.

4.3 Word Analysis of Negative Reviews

Since negative reviews always involve more important information, we have tackled them independently. At first, we collected all the “recommend” reviews with negative emotional tendencies and all the “not recommend” reviews. All these reviews are treated as negative reviews. The part-of-speech tagging and word frequency statistics are carried out to analyze the words with apparent changes in word frequency ranking among nouns, verbs and adjectives.

Words typically related to motion sickness increase significantly. At the same time, words related to the gameplay decrease. Verbs related to problems in the game, like “crash” and “break”, increase. Among adjectives, words that rise significantly in the ranking are mostly derogatory.

4.4 Dependency Parsing of High-frequency Words

The compound words of “bug” are significantly high, indicating that visual, physical, and other kinds of problems of bugs encountered by players will stop the game process and destroy the game experiences, which may be the reason for the dissatisfaction of players.

We then analyze the adjectives modifying “gameplay”. The top five most frequent adjectives show that players pay more attention

to the core gameplay of VR games, which is an essential part of the game that attracts players. Through the analysis of adjectives that modify “movement”, many VR players wish to have a free, smooth movement experience in VR games, but this is often accompanied by the generation of dizziness for players who cannot adapt to it. As can be seen from players’ reviews, the expectation of freely moving in VR games is intense. How to enhance the movement experience while reducing the symptoms of vertigo will be a real challenge for VR game developers.

5 CONCLUSION AND FUTURE WORKS

To improve and optimize the VR game experiences, we have studied 288,685 reviews from 506 VR games on the Steam platform and examined the relationship between the sentiment tendency and the recommended attitudes of reviews. Even though many players recommend a VR game, we find that they strictly put forward their discontent or further demands through reviews. It provides a new perspective for game review researchers as some of the recommended reviews also contain negative opinions about games. These reviews can give a larger sample and more apparent targets for user researches when ameliorating VR games. In addition, we find that players tend to pay more attention to movement and gameplay in VR games when they write reviews. The cybersickness that VR may cause is also a significant concern. The dependency parsing analysis shows that players pay more attention to the core gameplay, and they hope to have a free and smooth movement experience in VR games.

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